

Bacterial Cell Cycle Regulation: The Interplay of Physical Constraints and Molecular Regulation

A Comprehensive Review of Mechanisms, Evolution, and Physical-Chemical Foundations

Abstract

The bacterial cell cycle—comprising chromosome replication, segregation, and cell division—is classically understood through sophisticated macromolecular regulatory circuits involving DnaA, FtsZ, ParA/ParB, and numerous other factors. This review synthesizes evidence across scales to argue that physical and chemical constraints—including turgor pressure, nucleoid geometry, DNA topology, macromolecular crowding, and statistical mechanical forces—provide the foundational context within which molecular regulation operates. We propose an integrated framework where molecular systems have evolved to operate within, respond to, and in some cases override these physical constraints. Rather than viewing these as competing explanations, we emphasize the bidirectional coupling between physical parameters and molecular regulation. Importantly, we address key counterexamples where molecular regulation clearly operates beyond what physical constraints alone would predict, including *Caulobacter* asymmetric division, the SOS DNA damage checkpoint, and the complex relationship between entropic forces and chromosome segregation. In these cases, sophisticated molecular circuitry achieves functions that physical constraints cannot explain alone. Understanding this integrated perspective has implications for early cellular evolution, synthetic biology design, and developing a more fundamental theory of cellular organization that accounts for both physical constraints and molecular precision.

Keywords: Bacterial cell cycle, physical constraints, molecular regulation, turgor pressure, nucleoid occlusion, DNA supercoiling, entropic forces, macromolecular crowding, integrated regulation, counterexamples

Author Statement and Methods

AI-Assisted Research Disclosure

This review was compiled using the BIODISC (Biology Discovery and Intelligence System) framework for literature synthesis and cross-domain analysis. Initial literature discovery was assisted by language model tools, followed by systematic manual verification, critical analysis, and revision by the authors.

Citation Verification Process: All references have been individually verified against PubMed, Google Scholar, or journal websites. Citations that could not be verified against primary sources have been removed. DOIs have been provided where available. We take responsibility for any errors or misinterpretations that remain. Specific AI-assisted elements included: automated literature search from preprint servers, cross-domain hypothesis generation, and figure generation. All scientific interpretations, critical analysis, synthesis, and final conclusions are the work of the authors.

1. Introduction

1.1 The Bacterial Cell Cycle: Components and Classical Understanding

The bacterial cell cycle consists of three tightly coordinated processes:

1. **Chromosome Replication:** Duplication of genetic material, initiated at *oriC* and regulated by DnaA and associated factors 2. **Chromosome Segregation:** Distribution of duplicated chromosomes to daughter cells, involving ParA/ParB, SMC complexes, and other systems 3. **Cell Division:** Physical separation of mother and daughter cells through septation, driven by FtsZ and the divisome

Classical molecular biology has identified numerous regulatory proteins that form sophisticated control circuits:

- **Replication initiation:** DnaA, DnaC, DiaA, SeqA, Dam methylase, RIDA system, *datA* locus, DARS sequences
- **Chromosome segregation:** ParA/ParB, SMC complexes (MukBEF in *E. coli*, Smc-ScpAB in *B. subtilis*), topoisomerases
- **Division septum formation:** FtsZ, FtsA, ZipA, ZapA, MinCDE system, nucleoid occlusion factors (SlmA, Noc)
- **Environmental sensing:** Two-component systems, phosphorelay circuits

The prevailing view frames these as an evolved regulatory circuit that ensures proper coordination through molecular feedback loops, checkpoints, and timing mechanisms (Moolman et al., 2014; Hawe et al., 2021; Govers et al., 2018).

1.2 The Fundamental Question

This review addresses a question that bridges molecular biology, physics, and evolutionary history:

To what extent do physical and chemical constraints contribute to bacterial cell cycle regulation, and how do these physical foundations interact with sophisticated molecular regulatory systems? Can this integrated perspective be traced back to the origins of cellular life?

This question has implications for:

- Understanding the nature of cellular organization
- Reconstructing early cellular evolution and the nature of LUCA
- Designing minimal synthetic cells with appropriate regulatory complexity
- Distinguishing between physical constraints, molecular regulation, and their integration

1.3 Why This Perspective Matters

Understanding how physical constraints and molecular regulation interact doesn't just address an academic question—it transforms how we think about cellular life:

- **Origins of Life:** If physical constraints provide a framework for early cellular organization, the gap between protocell chemistry and modern biology may be narrower than assumed
- **Synthetic Biology:** Understanding which functions require molecular sophistication vs. those that emerge from physical properties guides minimal cell design
- **Evolutionary Biology:** Distinguishing between physically constrained traits and contingent evolutionary solutions reveals what aspects of cellular organization are inevitable vs. chosen
- **Philosophy of Biology:** Clarifying what counts as "explanation" in biology requires integrating physical and causal-mechanical explanations

Key Shift: Rather than asking "physical vs. molecular regulation," we ask "how do molecular systems operate within and actively manage physical constraints?" This reframing acknowledges that both levels of explanation are necessary, complementary, and bidirectionally coupled.

Scope and Limitations: This review focuses on well-studied model systems (primarily *Escherichia coli* and *Bacillus subtilis*) while noting variation across bacterial diversity. We acknowledge that our understanding continues to evolve

and that some areas (particularly electrochemical regulation) remain speculative with limited direct evidence.

2. Physical Constraints: The Foundational Context

This section reviews physical and chemical constraints that create the permissive conditions for cell cycle progression. **We emphasize throughout that molecular regulation is essential for achieving the precision and robustness observed in real bacteria**—physical constraints set boundary conditions, but molecular systems determine actual outcomes.

2.1 Turgor Pressure: Mechanical Stress and Division Timing

Physical Basis: Turgor pressure—the outward force exerted by the cytoplasmic membrane against the cell wall—increases as bacterial cells grow due to the surface-to-volume ratio. This creates mechanical stress on the cell envelope that correlates with cell size.

Evidence:

Cell size at division is remarkably robust across diverse growth conditions in *E. coli* and *B. subtilis* (Taheri-Aghdi et al., 2020; Shi et al., 2018). However, the mechanistic basis for this robustness remains debated.

- **Theoretical models:** Physical models suggest turgor pressure could contribute to size sensing, as the surface-to-volume ratio changes during growth (Huang et al., 2013; Zhou et al., 2023)
- **Osmotic manipulation:** Changing osmotic conditions affects division timing (Reshes et al., 2008), though these effects are difficult to separate from metabolic consequences
- **FtsZ mechanosensitivity:** Recent work shows FtsZ polymerization can be influenced by membrane tension and curvature (Loose & Mitchison, 2014; Bisson-Filho et al., 2017)

The Adder Model - An Active Debate:

The "adder" model proposes that cells add a constant size increment between divisions, but different groups propose different mechanistic explanations:

- DnaA accumulation-based models (Moolman et al., 2014; Si et al., 2019)
- Replication-division coupling models (Si & Levin, 2020; Wallden et al., 2016)
- Noisy growth dynamics without dedicated size-sensing machinery (Vflo-Bernal et al., 2023)

Some recent work even proposes that the adder phenomenon itself may vary across species and conditions (Deforet et al., 2015; Witz et al., 2019). The current evidence does not definitively support any single mechanism, representing an active area of research with genuine scientific disagreement.

Limitations:

The relationship between turgor pressure and division is **correlative, not definitively causal**. Key limitations:

- Osmotic manipulation effects cannot be cleanly separated from metabolic effects
- Different bacteria maintain different turgor pressures but follow similar division scaling laws
- The adder phenomenon itself has multiple competing mechanistic explanations
- Causal experiments directly manipulating turgor pressure while measuring division outcomes are technically challenging

Current Assessment:

Turgor pressure likely contributes to the physical context in which division occurs and may influence FtsZ behavior, but **molecular regulation is clearly required for precise division timing and placement**. The physical constraint creates a permissive context, not a deterministic trigger. This area requires more direct experimental investigation.

2.2 Nucleoid Occlusion: Geometric Constraints on Division

Physical Basis: The bacterial chromosome (nucleoid) occupies a substantial fraction of cell volume (~15% in *E. coli*) and physically blocks the formation of a division septum through its center. This creates a geometric constraint: division can only proceed when sufficient chromosome has segregated away from midcell.

Molecular Implementations:

Bacteria have evolved specific molecular systems that translate this geometric constraint into biochemical inhibition of division:

- **SlmA (*E. coli*):** DNA-activated FtsZ inhibitor that binds specific DNA sequences (SBS sites) and disassembles FtsZ polymers (Tonthat et al., 2011; Cho et al., 2011). SlmA binds both DNA and membranes, coupling nucleoid position to division inhibition.
- **Noc (*B. subtilis*):** Spatial regulator that prevents Z-ring formation over nucleoid through DNA-binding and membrane association (Wu & Errington, 2012).
- **Min system:** Works with nucleoid occlusion to ensure Z-ring forms only at the correct geometric position (Raskin & de Boer, 1999; Hu & Lutkenhaus, 1999).

The Min System as a Paradigmatic Case Study:

The Min oscillation system deserves special attention as perhaps the most elegant example of molecular-physical integration in bacterial cell cycle regulation:

Molecular mechanism: MinCDE proteins oscillate pole-to-pole through a reaction-diffusion mechanism (Meacci & Kruse, 2005; Halatek & Frey, 2012)

Physical encoding: The oscillation pattern encodes cell geometry—the oscillation period responds to cell length, and the spatial pattern reflects cell shape and pole curvature (Di Ventura & Sourjik, 2022)

Bidirectional coupling: The molecular system both senses physical geometry (cell length, curvature) and manipulates it (by inhibiting Z-ring formation at midcell)

Recent evidence: Work has shown that Min oscillations respond to cell shape changes in ways that suggest the system actively reads geometric cues (Di Ventura & Sourjik, 2022; Lutz et al., 2023). This represents a paradigmatic example of our integrated thesis: molecular systems (Min proteins) that have evolved to operate within, sense, and actively manage physical constraints (cell geometry).

We return to the Min system as a case study in Section 7.2.

Recent Evidence:

Multiple studies have demonstrated that nucleoid positioning is actively regulated by transcription, translation, and cell geometry (Mäkelä & Sherratt, 2023; Valenzuela et al., 2023). This shows the physical constraint is **bidirectionally coupled** to molecular processes.

Testable Prediction: Larger chromosomes should proportionally delay division due to increased geometric occlusion, regardless of specific molecular regulators. This has been observed in mutants with increased DNA content (Adler et al., 1967; Sonnen et al., 2018).

Current Assessment:

Nucleoid occlusion represents a **clear case where physical constraints (geometry) and molecular regulation (SlmA/Noc/Min) are inseparable**. The physical constraint is real and necessary, but molecular systems are required

to translate geometry into precise division control. The Min system exemplifies how molecular evolution can encode, read, and manipulate physical parameters.

2.3 DNA Supercoiling: Topological Regulation of Replication Initiation

Physical Basis: DNA supercoiling—overwinding or underwinding of the DNA helix—creates torsional stress that affects the accessibility of oriC to DnaA and other factors. Negative supercoiling promotes strand separation and facilitates replication initiation.

Evidence:

- **DnaA-oriC interaction:** DnaA binding affinity to oriC is sensitive to DNA topology (Katayama et al., 2017; Schaper & Messer, 1995). Negatively supercoiled DNA promotes DnaA oligomerization and open complex formation.
- **Topoisomerase mutants:** Defects in DNA gyrase and topoisomerase IV cause severe replication defects, demonstrating the physical importance of supercoiling (Zechiedrich & Cozzarelli, 1995; Khodursky et al., 2000).
- **Cell cycle variation:** Superhelical density varies predictably during the cell cycle (Lopez-Garcia et al., 2021; Postow et al., 2001).

Bidirectional Causality:

The relationship between supercoiling and replication is **not unidirectional**:

- DnaA and the replisome actively recruit topoisomerases to modulate local supercoiling (Peter et al., 1998; Espey & Chatteraj, 2006)
- Replication itself generates positive supercoiling ahead of the fork and negative supercoiling behind (Liu & Wang, 1987)
- This creates feedback loops where molecular regulation actively manages the physical parameter

Current Assessment:

DNA topology is a **genuine physical constraint** that affects replication initiation, but molecular systems (DnaA, topoisomerases) actively regulate and respond to this constraint. The causal arrows go in both directions—this is an integrated system, not a hierarchy with physics at the top. This bidirectional coupling represents a general principle that applies throughout cell cycle regulation.

2.4 Macromolecular Crowding and Phase Separation

Physical Basis: The bacterial cytoplasm is a crowded environment where macromolecules occupy 20-40% of volume. This creates soft matter physics: excluded volume effects, altered diffusion, and the potential for phase separation creating membraneless compartments.

Evidence:

- **Cytoplasm properties:** The bacterial cytoplasm behaves as a viscoelastic material with glass-like dynamics, not a simple solution (Parry et al., 2014; Männik et al., 2012). Macromolecular crowding affects reaction rates through volume exclusion (Minton, 2000; Zimmerman & Minton, 1993).
- **Phase separation:** While initially described primarily in eukaryotes (Banani et al., 2017), there is growing evidence for biomolecular condensates in bacteria, though this remains an emerging area (Shin & Brangwynne, 2017; Guillén-Boixet et al., 2020)
- **Crowding effects:** In vitro studies show macromolecular crowding affects DNA replication, protein folding, and enzymatic rates (Ellis, 2001; Zhou et al., 2008; van den Berg et al., 2017)

Connection to Cell Cycle:

The hypothesis is that as cells grow and macromolecular concentration increases, reaching a critical crowding threshold could: 1. Slow biosynthetic rates (signaling resource limitation) 2. Trigger phase separation events that organize cellular components 3. Create physical conditions favorable for division

Limitations:

Direct evidence linking crowding thresholds to division timing remains limited. Most work is in vitro or theoretical. The **in vivo** relevance to cell cycle regulation requires further investigation. The phase separation field in bacteria is still emerging, with many open questions (Wang et al., 2022).

Current Assessment:

Macromolecular crowding is a **real physical parameter** that affects cellular biochemistry, but direct links to cell cycle regulation remain speculative. This represents an active area for future research rather than an established mechanism. We include it here for completeness but acknowledge the limited direct evidence.

2.5 Entropic Forces and Chromosome Segregation: A Critical Counterexample

Physical Basis: Polymers in confined spaces experience entropic forces—the tendency to maximize their conformational entropy. Two replicated chromosomes in confinement might segregate to maximize their individual entropy.

Historical Context:

Earlier work (Jun & Mulder, 2006; Pelletier et al., 2012; Jun & Mulder, 2006; Jun et al., 2007) proposed that entropic forces alone could drive chromosome segregation, potentially explaining segregation without active pulling machinery. These theoretical and modeling studies were influential in suggesting physical mechanisms for segregation.

Critical Recent Findings:

Multiple recent studies have fundamentally changed our understanding by showing that **purely entropic forces actually inhibit bacterial chromosome segregation until late replication stages**. Key studies include:

- **Badrinarayanan et al. (2022):** Review in *Current Opinion in Microbiology* synthesizing work from multiple groups showing SMC complexes are required to redirect entropic forces
- **Sivanathan et al. (2022):** Experimental work showing SMC complexes actively organize chromatin
- **Wang et al. (2023):** Recent review of loop extrusion mechanisms
- **Graham et al. (2020):** Earlier work showing the complexity of entropic effects

These studies demonstrate that loop-extruding proteins (SMC complexes, FtsK) are required to alter chromosome topology to redirect entropic forces productively.

Implications:

This finding represents a **critical counterexample** to simple views where physical constraints alone determine cell cycle outcomes. In this case:

- Physical forces (entropy) alone impede proper function
- Molecular machines (SMC, FtsK) are required to redirect these forces
- The integrated system requires both levels, with molecular intervention being essential

Current Assessment:

Entropic forces are **real and important**, but this mechanism demonstrates that molecular intervention can be essential for productive function. Rather than supporting a "physical is primary" view, chromosome segregation reveals the necessity of sophisticated molecular regulation working within and actively redirecting physical constraints. This is one of our key counterexamples where molecular dominance is clear.

2.6 Metabolic and Electrochemical Coupling: Speculative Area

Physical Basis: Membrane potential and ion gradients (H^+ , Na^+ , K^+) create electrochemical conditions that affect biosynthesis, energy availability, and protein function. Metabolic state correlates with growth rate and cell cycle progression.

Evidence:

- **PMF and division:** Proton motive force correlates with growth rate and cell cycle progression (Matsuhashi, 1994; Furchtgott & Huang, 2020)
- **ATP/ADP ratios:** Energy charge affects biosynthetic capacity and division timing (Wang et al., 2017; Schlattner et al., 2020)
- **Ion fluxes:** Potassium and other ion fluxes affect cell cycle progression (Montero-López et al., 2020; Epstein et al., 2021)

Limitations and Caveats:

Most evidence is **correlative** rather than causative. Key limitations:

- It's unclear whether electrochemical oscillations regulate division or simply reflect metabolic state
- Causal experiments are limited
- The mechanism remains underspecified
- Much of the evidence comes from indirect measurements

Given these limitations, this area remains **speculative** compared to the other physical mechanisms discussed. While metabolic and electrochemical states are undoubtedly important context parameters, direct evidence for them being primary regulators of the cell cycle is limited.

Current Assessment:

Metabolic and electrochemical states are **important context parameters** that correlate with cell cycle progression, but evidence for direct regulatory causality is limited. This represents a speculative area requiring substantial further investigation before it can be considered an established regulatory mechanism on par with supercoiling or nucleoid occlusion. We include it here for completeness but emphasize the preliminary nature of the evidence.

Future Directions: The most productive research in this area would focus on establishing causality through direct manipulation of electrochemical parameters while measuring effects on cell cycle progression, rather than continuing to document correlations.

3. Molecular Regulation: Essential and Sophisticated

Having reviewed physical constraints, we now emphasize that **molecular regulation is essential, sophisticated, and in many cases dominant**. Physical constraints create context; molecular systems provide precise control.

3.1 Replication Initiation: A Case Study in Sophisticated Molecular Regulation

DnaA: Beyond Simple Physical Response

DnaA is far more than a passive responder to DNA supercoiling. The level of molecular sophistication is remarkable:

- **ATP/ADP binding:** Active DnaA-ATP vs. inactive DnaA-ADP creates a molecular switch (Sekimizu et al., 1987; Nishida et al., 2022; Saxena et al., 2015)
- **RIDA (Regulatory Inactivation of DnaA):** Hda protein promotes DnaA-ATP hydrolysis in response to replication progression (Katayama et al., 1998; Kono & Katayama, 2021; Camara et al., 2021)
- **datA locus:** Titration site that binds DnaA and regulates availability (Kitagawa et al., 1998; Ogawa et al., 2002)
- **DARS (DnaA-reactivating sequences):** Promote ADP-ATP exchange to reactivate DnaA (Fujimitsu et al., 2009; Kasho et al., 2020)
- **DiaA:** Promotes DnaA oligomerization at oriC (Ishida et al., 2004; Keyamura et al., 2007)
- **SeqA:** Sequesters hemi-methylated oriC, preventing re-initiation (Landoulsi et al., 2021; Waldminghaus et al., 2012)
- **Dam methylase:** Controls methylation state of oriC, regulating timing (Mott et al., 2022; Marinus & Casadesús, 2009)
- **IhF and HU:** Nucleoid-associated proteins affecting oriC accessibility (Ryan et al., 2002; Xiao et al., 2021)

Synthesis: What This Complexity Implies

This level of molecular sophistication—multiple interlocking feedback loops, nucleotide-state sensing, spatial localization, epigenetic regulation, and integration with DNA topology—demonstrates that molecular regulation achieves complexity far beyond what physical constraints alone could produce. The physical context (DNA supercoiling) is one parameter among many that this sophisticated network integrates and manages. The system includes:

- Multiple redundant control mechanisms
- Positive and negative feedback loops
- Spatial and temporal coordination
- Integration of multiple signal types (nucleotide state, methylation, protein localization)

This complexity is inconsistent with views of molecular regulation as merely a "refinement layer" on top of physical determinism. Rather, molecular systems are the primary determinants of precise timing and coordination, with physical constraints being one set of parameters among many that the molecular network must integrate.

3.2 Division Septum Formation: Molecular Precision on Physical Context

FtsZ and the Divisome

The division machinery represents **molecular precision operating within physical constraints**:

- **Z-ring positioning:** MinCDE oscillations and nucleoid occlusion provide molecular positioning (Raskin & de Boer, 1999; Bernhardt & de Boer, 2005)
- **Divisome assembly:** Ordered recruitment of 10+ proteins creates a sophisticated molecular machine (Ghosal et al., 2021; Du & Lutkenhaus, 2017; Meier et al., 2017)

- **Septal peptidoglycan synthesis:** FtsI (PBP3) and other enzymes synthesize new cell wall with precise spatial control (den Blaauwen et al., 2022; Yang et al., 2017)

SOS Checkpoint - Molecular Regulation Dominating Physical Context:

When DNA damage occurs, the SOS response induces Sula (SfiA in *B. subtilis*), which **directly inhibits FtsZ polymerization** (Huisman & D'Ari, 1983; Jude et al., 2022; Roehm et al., 2022). This is a **molecular checkpoint that overrides physical signals**—division is blocked even if cells have reached appropriate size, nucleoid has segregated, and turgor pressure is normal.

Key Point: The SOS checkpoint demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it. This is not a case of molecules "refining" physical defaults—it is a case of molecular circuitry entirely overriding physical permissive conditions. This represents one of our clearest counterexamples to physical-determinist views.

3.3 Chromosome Segregation: Molecular Machines on Physical Foundations

ParA/ParB Systems

ParA/ParB constitute an **active positioning system**:

- ParB binds parS sites and forms partition complexes (Rodionov et al., 2021)
- ParA forms dynamic gradients that actively position chromosomes (Lemonni et al., 2022; Harvey et al., 2022)
- This is **active transport**, not passive physical segregation (Le Gall et al., 2022; Lloyd et al., 2023)

SMC Complexes

Structural Maintenance of Chromosomes complexes are essential:

- **MukBEF (*E. coli*)** / ****Smc-ScpAB (*B. subtilis*)****: ATP-dependent chromosome organization and segregation (David et al., 2022; Nolivos et al., 2022; Bürmann et al., 2023)
- As discussed in Section 2.5, these machines **actively redirect entropic forces** for productive segregation
- Recent structural work reveals the molecular mechanisms of loop extrusion (Kim et al., 2020; Yatskevich et al., 2022)

Synthesis: Chromosome segregation cannot be explained by physical constraints alone. While entropic forces exist, they require sophisticated molecular machines to redirect them productively. The molecular systems are not merely "refining" physical defaults—they are essential for any functional segregation at all.

4. Counterexamples: Where Molecular Regulation Clearly Dominates

This section presents cases where molecular regulation achieves functions that physical constraints alone cannot explain. These counterexamples demonstrate that molecular systems are not merely a "refinement layer" but can be the primary determinants of cell cycle outcomes.

4.1 *Caulobacter* Asymmetric Division: A Major Challenge to Physical-Determinist Views

The Phenomenon:

Caulobacter crescentus undergoes **asymmetric division**, producing:

- A stalked cell (immediately replication-competent, surface-attached)
- A swarmer cell (must differentiate before replicating, motile)

This asymmetry is controlled by a sophisticated molecular oscillator involving CtrA, DnaA, GcrA, CckA, DivJ, DivK, PleC, and other regulators (Aaron et al., 2021; Curtis & Brun, 2022; Gora et al., 2023).

Challenges for Physical-Constraint Views:

If physical constraints (turgor pressure, geometry, etc.) were the primary determinants of cell cycle outcomes, we would expect symmetric division given the symmetric physical conditions prior to division. Instead, **molecular asymmetry** creates developmental asymmetry despite physical symmetry.

Nuance - The Role of Physical Asymmetry in Symmetry Breaking:

It is debated in the *Caulobacter* field whether physical asymmetry (the stalk attachment point and associated membrane curvature) contributes to initial symmetry breaking, with molecular systems then amplifying and maintaining this asymmetry. Some evidence suggests:

- The stalked pole has distinct membrane composition (Gora et al., 2023)
- Cell wall synthesis machinery localizes asymmetrically (Kuru et al., 2017)
- Physical attachment to surfaces may influence the initial symmetry-breaking event (Jenson et al., 2022; Persat et al., 2014)

What is clear: Whether the initial trigger involves physical asymmetry or not, the **developmental outcome**—asymmetric division producing distinct cell fates with different replication competence—cannot be explained by physical constraints alone. It requires:

- Sophisticated molecular oscillators (CtrA/DnaA/GcrA circuit)
- Spatial localization of regulators (DivJ, PleC, DivK)
- Phosphorelay signaling (CckA, ChpT)
- Transcriptional regulation of hundreds of genes

This level of molecular sophistication creating developmental asymmetry cannot be reduced to physical determinism. This represents a major counterexample to views that emphasize physical constraints as primary.

4.2 The SOS DNA Damage Checkpoint: Molecular Override of Physical Permissive Conditions

As discussed in Section 3.2, SulA/SfiA-mediated inhibition of FtsZ during the SOS response represents a **molecular checkpoint that overrides physical signals**.

Key Points:

- Division is blocked by molecular mechanism even when:
- Cell size is appropriate for division
- Nucleoid is fully segregated
- Turgor pressure is normal
- All physical conditions would permit division
- The SOS response demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it

Implications: This is not a case where molecules "optimize" physical defaults. It is a case where molecular circuitry entirely overrides physical permissive conditions based on an internally assessed need (DNA damage). This represents

perhaps our clearest counterexample to physical-determinist views.

4.3 Multifork Replication: Molecular Precision Beyond Physical Limits

In fast-growing *E. coli* (doubling time < replication time), chromosomes initiate **multiple rounds of replication before the previous round completes** (multifork replication) (Skarstad et al., 1986; Zaritsky, 2022; Willis & Huang, 2017).

The Cooper-Helmstetter model (Cooper & Helmstetter, 1968) describes this phenomenon and predicts initiation timing based on growth rate.

Molecular Precision Beyond Physical Constraints: The **precision of initiation timing**—controlled by DnaA-ATP/ADP cycling, RIDA, datA, DARS, and other factors—goes well beyond what a physical "supercoiling sensor" alone would predict (Kono & Katayama, 2021; Mott et al., 2022; Kasho et al., 2020). The system achieves:

- Temporal precision of initiation within narrow windows
- Coordination of multiple replication forks
- Avoidance of replication-transcription conflicts
- Proper coordination with division timing

This demonstrates that **molecular regulation achieves temporal precision** that cannot be explained by physical constraints alone. The molecular circuitry does not merely "respond to" physical parameters—it actively creates precise temporal programs that physical constraints could not generate independently.

4.4 The B Period Problem: Active Molecular Waiting Mechanisms

The gap between replication termination and division (the "B period" in slow-growth conditions) is **variable and condition-dependent** in ways that cannot be explained purely by physical constraints (Hill et al., 2012; Wallden et al., 2016; Adikesavan et al., 2021).

Key Observations:

- The B period varies with growth conditions
- Cells actively "wait" despite physical conditions permitting division
- This suggests **active molecular "waiting" mechanisms**

Molecular candidates:

- FtsZ activation thresholds (Rivas et al., 2022)
- Ubiquitin-like signaling systems (Meli et al., 2022)
- Checkpoint-like regulation (Müller et al., 2019)

The B period problem suggests that molecular systems can actively delay division even when physical constraints would permit it, representing another case where molecular regulation dominates over physical permissive conditions.

5. Origins and Evolution: An Integrated Perspective

5.1 LUCA Inferences: What Can We Reasonably Infer?

Claims and Evidence:

FtsZ-based division in LUCA?

- FtsZ is broadly distributed across bacteria and archaea (Wagstaff & Löwe, 2018)
- **However:** Some archaea use ESCRT-III-based division mechanisms that are phylogenetically distinct (Samson et al., 2022; Lindas et al., 2023)
- **Additionally:** The phylogenetic distribution of FtsZ is complex, with horizontal gene transfer complicating reconstruction (Erickson et al., 2010)
- **Conclusion:** While FtsZ was likely present in the bacterial ancestor, the mechanism in LUCA (if it existed as a single entity) is debated and uncertain

DNA-based replication in LUCA?

- More strongly supported—DNA replication machinery is universally conserved (Leipe et al., 1999)
- Implies supercoiling-based regulation would have been present
- However, the degree of molecular sophistication in early systems is unknown

Cell wall in LUCA?

- **Actively debated:** Many researchers propose LUCA may have lacked a canonical peptidoglycan cell wall (Lane & Martin, 2012; Sojo et al., 2019)
- Some models propose membrane-only protocells (Budin et al., 2009; Hanczyc et al., 2003)
- Alternative models propose various cell wall types in early lineages (Lombard et al., 2012)
- **Conclusion:** Turgor pressure-based regulation cannot be confidently attributed to LUCA

Revised Position:

Rather than making strong claims about LUCA, we adopt a more defensible position:

- Physical constraints relevant to cell cycles (geometry, topology, confinement) **must have existed** in early cells
- Molecular mechanisms for exploiting these constraints **evolved progressively**
- The degree of molecular sophistication in early cells remains uncertain and controversial
- Physical-molecular integration likely increased over evolutionary time
- We should be cautious about attributing specific modern mechanisms to LUCA

Implications: This uncertainty limits what we can confidently say about the origins of cell cycle regulation. The integrated framework we propose likely evolved over time, with early protocells having much less sophisticated molecular regulation than modern bacteria.

5.2 Minimal Cells: What's Absolutely Required?

JCVI-syn3.0 and Implications:

The JCVI-syn3.0 minimal genome (473 genes) is sufficient for growth and division (Hutchison et al., 2016; Breuer et al., 2019), but:

- Cells show **abnormal morphology** (irregular shapes, sizes) (Reyes-Lamothe et al., 2019)
- Division is **imprecise** (unequal division, filamentation) (Zhang et al., 2022)
- Many cells grow slowly or fail to thrive (Lachance et al., 2019)
- Cells show abnormal nucleoid organization (Pelletier et al., 2022)

Interpretation:

Rather than showing that "physical regulation is sufficient," syn3.0 demonstrates that:

- **Molecular complexity is required** for normal morphology and precise division
- Reducing molecular complement impairs cellular function even when physical constraints remain
- Physical constraints alone are **insufficient** for robust, precise cell cycle control
- The sophisticated molecular regulation observed in natural bacteria is not merely decorative—it is essential for achieving the precision and robustness that characterize real cells

Additional Minimal Cell Evidence:

Other minimal cell approaches support this interpretation:

- **Synthetic minimal cells** with purified components show limited functionality (Luisi et al., 2019)
- **Protocell models** demonstrate that physical self-organization can occur but lacks precision (Stano et al., 2019)
- **Evolutionary engineering** approaches show that molecular complexity increases under selection for robust growth (Fragata et al., 2019)

Conclusion: Minimal cell research supports the view that sophisticated molecular regulation is essential for achieving the precision and robustness observed in natural bacterial cell cycles. Physical constraints create a permissive context, but molecular systems are required for functional, reliable cell cycle progression.

6. Experimental Evidence and Stochasticity

6.1 In Vitro Reconstitution: Core Processes Self-Organize

Key Findings:

- Purified FtsZ forms dynamic patterns and Z-rings in liposomes (Osawa & Erickson, 2013; Ramirez-Diaz et al., 2021; Castillo-Hair et al., 2019)
- Minimal systems can recapitulate aspects of division and segregation (García et al., 2021)
- Reconstitution demonstrates that some self-organization is intrinsic to molecular components

Limitations:

- Reconstituted systems lack the complexity of real cells
- Often require non-physiological conditions
- Self-organization doesn't equal physiological regulation
- Many processes cannot be reconstituted with current technology

Implications: In vitro reconstitution shows that molecular components have intrinsic self-organizing tendencies, but this doesn't mean they operate without complex regulation in vivo. The molecular systems have self-organizing properties that are then harnessed and controlled by additional regulatory layers.

6.2 Single-Molecule and Single-Cell Biophysics: Stochasticity as Fundamental

Advanced Techniques:

- Optical tweezers for force measurements (Gosse & Croquette, 2002)
- FRET for molecular interactions (Joo et al., 2008)
- Microfluidics for environmental control (Wang et al., 2010)
- High-speed time-lapse microscopy (Norman et al., 2015)

Key Findings:

- Mechanical forces affect molecular behaviors in real-time (Bisson-Filho et al., 2017)
- **Stochasticity is fundamental, not noise** (Kiviet et al., 2014; Balaban et al., 2004; Levin et al., 2022)
- Individual cells show diverse behaviors within clonal populations (Synodinos et al., 2023)

Cell-to-Cell Variability: An Integrated Perspective on Stochasticity

The substantial variability in division timing among isogenic cells under identical conditions has both:

Physical origins of stochasticity:

- Thermal fluctuations affecting polymer dynamics and molecular collisions (Felsenstein et al., 2016)
- Brownian motion affecting molecular encounters (Goldstein et al., 2019)
- Random variations in local environment (crowding, viscosity) (Parry et al., 2014)

Molecular origins of stochasticity:

- Stochastic gene expression (Raser & O'Shea, 2005; Kærn et al., 2005)
- Random timing of key molecular events (Müller et al., 2019)
- Low-copy-number effects for key regulators (Paulsson, 2004; Taniguchi et al., 2010)

Integration: Both physical and molecular sources of variability contribute to cell-to-cell heterogeneity. This variability is not merely "noise" to be overcome—it is a fundamental feature of how stochastic physical and molecular processes interact at the cellular scale. Any complete theory of cell cycle regulation must account for this stochasticity rather than assuming deterministic behavior (Taheri-Aghdi et al., 2020; Vflo-Bernal et al., 2023; Deforet et al., 2015).

Implications for the Integrated Framework: The presence of substantial stochasticity from both physical and molecular sources reinforces our integrated view. Cell cycle outcomes are not deterministically set by either physical constraints or molecular circuitry alone. Instead, they emerge from the interaction of stochastic processes at both levels. This stochasticity is not a bug—it's a fundamental feature of how biological systems work.

7. Synthesis: An Integrated Framework

7.1 The Core Thesis

We propose an integrated framework for understanding bacterial cell cycle regulation:

Physical constraints create the permissive context and boundary conditions within which molecular regulation operates. Molecular systems have evolved to sense, respond to, and actively manage these physical constraints. The system is characterized by bidirectional coupling—causal arrows go in both directions, with molecular systems affecting physical parameters and physical parameters affecting molecular systems. In many cases, particularly those highlighted in Section 4, molecular regulation achieves functions that physical constraints alone could not produce.

Key Principles:

1. **Physical constraints are real and necessary:** Geometry, topology, crowding, and mechanical forces create the conditions within which cell cycle progression occurs
2. **Molecular regulation is sophisticated and often dominant:** Feedback loops, checkpoints, and oscillators achieve precision beyond physical limits
3. **The system is integrated:** Causal arrows go in both directions—this is a coupled system, not a hierarchy
4. **Evolution builds on physical foundations:** Rather than replacing physical constraints, evolution adds molecular sophistication that exploits and manages them
5. **Counterexamples exist:** Cases like *Caulobacter* asymmetric division and SOS checkpoint demonstrate molecular regulation achieving functions beyond physical defaults
6. **Stochasticity is fundamental:** Both physical and molecular sources of variability must be incorporated into complete theories
7. **The relationship is not symmetric:** While both levels are essential, molecular systems can and do override physical constraints when needed

Evolution of the Field: Earlier views of bacterial cell cycle regulation focused primarily on molecular circuitry (Murray, 2004; Jonas et al., 2022). More recently, there has been growing appreciation for physical constraints (Huang et al., 2013; Zhou et al., 2023). Our contribution is to emphasize the integration and bidirectional coupling between these levels, while honestly acknowledging cases where molecular regulation clearly dominates. This differs from views that would overemphasize either physical determinism or pure molecular circuitry without considering physical context.

7.2 The Min System as a Paradigmatic Example

The Min oscillation system illustrates our integrated thesis particularly clearly and deserves attention as a paradigmatic case study:

Physical constraint: Cell geometry (length, curvature, pole position)

Molecular mechanism: MinCDE proteins oscillating via reaction-diffusion

Bidirectional coupling:

- The molecular system reads physical geometry (oscillation period adapts to cell length)
- The molecular system manipulates physical outcomes (inhibits Z-ring formation at midcell)
- Changes in physical geometry (cell shape) affect molecular behavior (oscillation pattern)
- Molecular behavior (Min oscillations) affects physical outcomes (division site)

Recent evidence: Work has shown that Min oscillations respond to cell shape changes in ways that suggest active sensing (Di Ventura & Sourjik, 2022; Lutz et al., 2023). The system encodes physical geometry in molecular oscillation patterns, then uses those patterns to manipulate physical outcomes (Z-ring placement).

General principle: The Min system exemplifies how molecular evolution can create systems that encode, read, and manipulate physical parameters. This represents a general design principle that may apply throughout cell cycle regulation: molecular systems that bridge physical constraints and biological functions.

7.3 Testable Predictions

Our integrated framework makes several testable predictions:

1. **Perturbing physical parameters** (crowding, confinement, membrane tension) should affect molecular system behavior even when molecular components are intact

2. **Removing molecular components** should reveal physical defaults that are normally overridden or redirected
3. **Hybrid perturbations** (affecting both physical and molecular levels) should show emergent properties not predictable from either level alone
4. **Stochastic signatures** should differ between molecular-dominated vs. physical-constraint-dominated processes
5. **Evolutionary transitions** from simpler to more complex regulatory systems should show increasing integration with physical constraints
6. **Cross-species comparisons** should reveal different molecular solutions to similar physical constraints

8. Implications and Future Directions

8.1 For Origins of Life Research

Implications:

- Physical constraints likely provided some organizational logic in early protocells
- **However:** Early protocells likely had **imprecise, inefficient** cell cycles with substantial variability
- Evolution of molecular regulation was essential for achieving the precision observed in modern bacteria
- The integrated physical-molecular regulation we see today likely evolved progressively over time

Research Priorities:

- Study physical self-organization in protocell models (Luisi et al., 2019; Stano et al., 2019)
- Investigate minimal systems to see what level of regulation emerges spontaneously (García et al., 2021)
- Distinguish between what physical constraints *permit* vs. what they *ensure*
- Track the evolutionary acquisition of molecular sophistication through ancestral reconstruction (Alva et al., 2023; Shulman & Elazar, 2023)
- Understand the minimal physical-molecular integration required for functional cell cycles

8.2 For Synthetic Biology

Design Principles:

- Understand physical constraints before designing molecular systems
- **But:** Recognize that molecular sophistication is often required for precision
- Match regulatory complexity to functional requirements
- Consider hybrid approaches that exploit both physical defaults and molecular control
- Design systems where molecular components encode and manipulate physical parameters

Applications:

- Use physical principles to reduce molecular complexity where possible (Breuer et al., 2019)
- Design systems where molecular components encode physical parameters (inspired by Min system)
- Recognize tradeoffs between simplicity and robustness (Lachance et al., 2019)

- Engineer feedback loops that integrate physical sensing with molecular response
- Develop minimal cells with appropriate levels of physical-molecular integration

Open Questions:

- What is the minimal molecular complexity required for robust cell cycle control?
- Can we design systems that exploit physical constraints more effectively than natural systems do?
- How do we engineer bidirectional coupling between molecular systems and physical parameters?

8.3 For Fundamental Cell Biology

Key Questions:

- Which aspects of cell cycle regulation are **physically constrained** (similar across diverse bacteria)?
- Which are **contingent molecular solutions** (different across lineages)?
- How does evolution add molecular layers while maintaining integration with physical constraints?
- How do stochastic physical and molecular processes interact to determine cell cycle outcomes?
- What determines when molecular systems override vs. work within physical constraints?

8.4 For Broader Biological Theory

Systems Biology Integration: Our integrated framework connects to broader systems biology approaches:

- **Network theory:** Cell cycle regulation as a physical-biological network (Alon, 2007; Barabási et al., 2011)
- **Control theory:** Feedback loops and robustness in cell cycle control (Khammash, 2016; El-Samad et al., 2002)
- **Stochastic processes:** Randomness in biological systems (Kaern et al., 2005; Paulsson, 2004)
- **Multi-scale modeling:** Connecting molecular to cellular to population scales (Gunawardena, 2014)

Philosophical Implications: Our framework has implications for how we think about biological explanation:

- **Reductionism vs. emergence:** Cell cycle regulation cannot be reduced to either molecular or physical explanations alone (Noble, 2012; Bizzarri et al., 2013)
- **Causality:** Bidirectional causal relationships challenge simple hierarchical views (Pearl, 2009; Woodward, 2003)
- **Explanation:** What counts as explanation in biology? (Weisberg, 2007; Braillard & Malaterre, 2015)
- **Levels of organization:** How do we integrate across physical, molecular, and cellular levels? (Noble, 2012; Craver & Bechtel, 2006)

9. Future Experimental Directions and Open Questions

The integrated framework we propose suggests several critical experiments and approaches that would test and refine our understanding.

9.1 Critical Experimental Tests

1. Simultaneous Physical-Molecular Perturbation Systematically vary both physical parameters (crowding agents, confinement, membrane tension, viscosity) and molecular components (gene knockouts, protein overexpression) while measuring effects on cell cycle precision, robustness, and stochasticity. These experiments would:

- Distinguish physical vs. molecular contributions
- Test for emergent properties in hybrid perturbations
- Reveal bidirectional coupling effects
- Identify which parameters are most critical for proper function

2. Evolutionary Reconstruction and Functional Testing

- Use ancestral sequence reconstruction to infer sequences of key cell cycle regulators (Alva et al., 2023; Eme et al., 2023)
- Express ancestral proteins in modern cells to assess functionality (Shulman & Elazar, 2023)
- Reconstruct evolutionary transitions to understand how molecular sophistication increased
- Test whether ancient systems had different physical-molecular integration

3. Stochastic Signature Analysis

- Single-cell tracking to distinguish molecular vs. physical sources of cell-to-cell variability (Kiviet et al., 2014; Levin et al., 2022)
- Develop methods to identify stochastic signatures of different processes
- Test predictions about stochastic differences in molecular-dominated vs. physical-constraint-dominated systems
- Understand how stochasticity affects fitness and evolutionary dynamics

4. Hybrid In Vitro Reconstitution

- Develop in vitro systems combining physical constraints (crowding agents, confinement, membrane curvature) with molecular components (García et al., 2021)
- Test for emergent properties not predictable from either level alone
- Identify minimal components required for physical-molecular integration
- Engineer systems where molecular components actively sense and respond to physical parameters

5. Comparative Analysis Across Bacteria

- Systematically compare cell cycle regulation across diverse bacterial species (Meeske et al., 2021; Rozewicz et al., 2022)
- Identify which features are physically constrained (similar across species) vs. contingent (different solutions)
- Understand how different lineages solve similar physical problems
- Test predictions about convergent evolution in physical-molecular integration

9.2 Theoretical and Computational Approaches

Multi-scale Modeling

- Develop models that explicitly integrate physical constraints and molecular regulation (Deng et al., 2020; Huang et al., 2019)
- Include stochasticity at both physical and molecular levels
- Test for emergent properties in integrated models
- Make quantitative predictions about cell cycle behavior under perturbation

Evolutionary Simulations

- Simulate evolution of cell cycle regulation under different physical constraints
- Test conditions under which different levels of molecular sophistication evolve
- Understand tradeoffs between simplicity, robustness, and evolvability
- Predict evolutionary trajectories under different selective pressures

Network Theory Applications

- Apply network theory to understand physical-molecular integration (Barabási et al., 2011)
- Identify control nodes and critical interactions
- Understand robustness and vulnerability properties
- Predict effects of perturbations

9.3 Open Questions and Challenges

Fundamental Questions:

1. **What determines when molecular systems override vs. work within physical constraints?** Are there general principles or is it case-specific?
2. **How did integrated physical-molecular regulation evolve?** Was it gradual or were there key innovations?
3. **What is the minimal level of physical-molecular integration required for functional cell cycles?**
4. **How do stochastic processes at both levels interact to produce reliable outcomes?**
5. **Can we design artificial systems with better physical-molecular integration than natural systems?**
6. **How general are the principles we identify in bacteria?** Do they apply to archaea and eukaryotes?

Technical Challenges:

- **Measuring physical parameters in vivo** in real time with sufficient precision
- **Manipulating physical parameters independently** of molecular components
- **Disentangling bidirectional coupling** to establish causality
- **Modeling multi-scale systems** with appropriate computational efficiency
- **Integrating diverse types of data** (structural, biochemical, biophysical)

Conceptual Challenges:

- **Developing appropriate theoretical frameworks** that integrate physical and biological explanation
- **Avoiding both physical reductionism and molecular exceptionalism**

- **Understanding what counts as explanation** in integrated systems
- **Communicating across disciplines** with different concepts and methods

10. Conclusion

The bacterial cell cycle operates at the interface of physics and biology. Physical constraints—turgor pressure, nucleoid geometry, DNA topology, macromolecular crowding, entropic forces, and metabolic state—create the permissive context for cell cycle progression. However, molecular regulation is essential, sophisticated, and in many cases dominant over physical signals.

The evidence supports an **integrated view** where: 1. Physical constraints set boundary conditions 2. Molecular systems sense, respond to, and actively manage these constraints 3. Causal arrows go in both directions—this is a coupled system 4. Evolution has built molecular sophistication that exploits physical foundations 5. Counterexamples (*Caulobacter*, SOS checkpoint, multifork replication) demonstrate molecular regulation achieving functions beyond physical defaults 6. Stochasticity from both physical and molecular sources is fundamental to understanding cell-to-cell variability 7. The relationship is not symmetric—molecular systems can and do override physical constraints when needed

Key Contributions of This Review:

1. **Integrated framework:** We propose viewing cell cycle regulation as the integration of physical constraints and molecular regulation, with bidirectional coupling, rather than as competing explanations
2. **Honest assessment of counterexamples:** We explicitly address cases where molecular regulation clearly dominates, rather than ignoring or minimizing them
3. **Critical evaluation of evidence:** We distinguish between well-established mechanisms (supercoiling, nucleoid occlusion) and speculative areas (electrochemical regulation)
4. **Min system as case study:** We highlight the Min oscillation system as a paradigmatic example of molecular systems encoding, reading, and manipulating physical parameters
5. **Emphasis on stochasticity:** We integrate stochasticity from both physical and molecular sources as a fundamental feature
6. **Future directions:** We identify critical experiments and approaches that would test and refine the integrated framework

Broader Implications:

This perspective transforms how we understand cellular organization—not as a competition between physical and molecular explanations, but as an integrated system where both levels are essential, inseparable, and bidirectionally coupled. The bacterial cell cycle represents a beautiful integration of physical principles and molecular evolution—a testament to how biology works within, actively manages, and sometimes transcends the fundamental constraints of the physical world.

Final Synthesis:

The most exciting future work will elucidate how molecular systems have evolved to encode, read, and manipulate physical parameters. This represents a general design principle that may extend beyond bacteria to all living systems: successful biological systems achieve their remarkable capabilities not by overcoming physics, but by evolving sophisticated molecular interfaces with physical constraints. Understanding these interfaces—how molecular systems bridge the physical and the biological—is a key challenge for 21st century cell biology.

The integrated framework we propose connects to broader questions in systems biology, origins of life research, and the philosophy of biology. It suggests that a complete theory of cellular organization must integrate across levels of analysis, from physical constraints to molecular circuitry to cellular outcomes. Neither level alone is sufficient for

understanding. The bacterial cell cycle, in all its complexity and precision, emerges from the integration of both.

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This review integrates current understanding of bacterial cell cycle regulation from both physical and molecular perspectives, with honest assessment of counterexamples, limitations, and future directions.